Closing the Loop

A Unified Approach to State-Space Kalman Decomposition, Reduced Order Luenberger Observer Design and Optimal LQR Control

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Introduction

In this assignment, we develop a unified state-space framework that integrates these core aspects into a coherent workflow, beginning with system analysis and culminating in closed-loop control and visualization.

We start with an examination of controllability and observability using the Popov–Belevitch–Hautus (PBH) tests, followed by Kalman decomposition to isolate the intersections of controllable, uncontrollable and observable, unobservable subspaces of the system and use them to construct the similarity transformation.

A reduced-order Luenberger observer is then constructed to estimate detectable states, ensuring asymptotic convergence of the estimation error. Next, an optimal state-feedback controller is designed using the Linear Quadratic Regulator (LQR) framework, balancing performance and control effort through the solution of the Algebraic Riccati Equation(ARE). These components are combined to form a complete output-feedback control law.

To validate the theoretical design, the closed-loop dynamics are analyzed through eigenvalue inspection and phase portrait visualization. The framework is further applied to a nonlinear benchmark system, the Furuta pendulum, where the effectiveness of the linearized control strategy is illustrated through animation. Finally, the impact of stochastic disturbances is introduced through additive white Gaussian noise, providing insight into robustness and practical performance limitations.

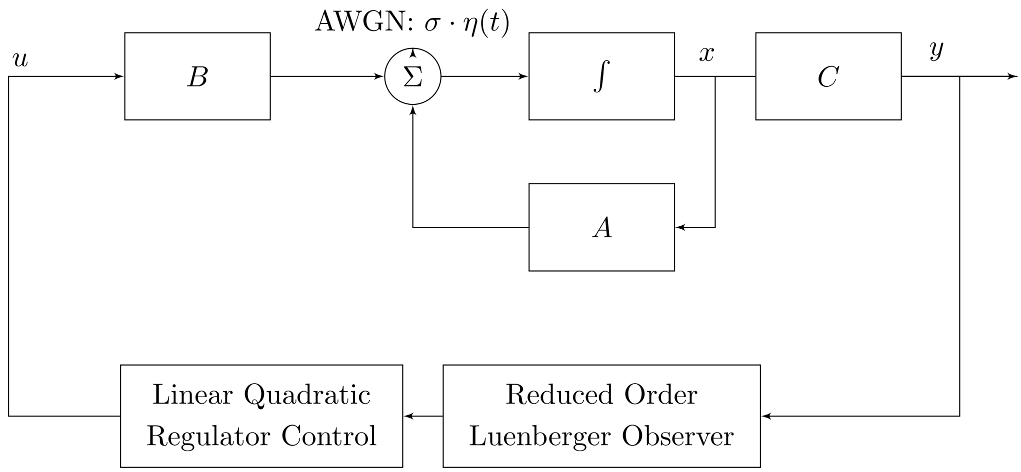
Aim

To develop and analyze a complete state-space control framework by:

- Evaluating system controllability and observability using PBH tests
- Performing Kalman decomposition to isolate $V_{co}, V_{\bar{c}o}, V_{\bar{c}\bar{o}}, V_{c\bar{o}}$ subspaces.
- Designing a reduced order Luenberger observer for state estimation by first finding the observer gain matrix(L) through Lyapunov stability using the Gramian matrix.
- Synthesizing an optimal state-feedback controller using the LQR approach to find the state feedback gain matrix (K).
- Integrating the Luenberger observer and LQR control into a closed loop output feedback system.

Theory

The system blocks can be defined as shown below:-



State-Space Representation

A linear time-invariant (LTI) system is described as:

$$\dot{x} = Ax + Bu, \quad y = Cx$$

where $x \in \mathbb{R}^{14}$ is the state vector, $u \in \mathbb{R}^3$ is the control input, and $y \in \mathbb{R}^2$ is the output.

Controllability and Observability

A system is controllable if any state can be reached using an input u. The PBH test states:

$$\text{rank}([\lambda I - A|B]) = n, \quad \forall \lambda \in \text{eig}(A)$$

Similarly, a system is observable if the state can be reconstructed from outputs:

$$\text{rank} \begin{bmatrix} A - \lambda I \\ C \end{bmatrix} = n, \quad \forall \lambda \in \text{eig}(A)$$

Kalman Decomposition

The state space is transformed using a similarity transformation:

$$\tilde{x} = T^{-1}x$$

yielding a block-structured system:

$$\tilde{A} = T^{-1}AT, \quad \tilde{B} = T^{-1}B, \quad \tilde{C} = CT$$

Reduced-Order Luenberger Observer

After decomposition, the system is partitioned as:

$$\begin{aligned}
\dot{x}_1 &= A_{11}x_1 + A_{12}x_2 + B_1u \\
\dot{x}_2 &= A_{21}x_1 + A_{22}x_2 + B_2u \\
y &= x_1
\end{aligned}$$

Since x_1 is measured, only x_2 is estimated using:

$$\dot{\hat{x}}_2 = A_{22}\hat{x}_2 + A_{21}y + B_2u + L(x_1 - \hat{x}_1)$$

The estimation error dynamics are:

$$\dot{e} = (A_{22} - LA_{12})e$$

The observer gain L is chosen such that $(A_{22} - LA_{12})$ is stable. This is achieved using the observability Gramian matrix:-

$$W_o = \int_0^\infty e^{A^T t} C^T C \, e^{At} \, dt$$

Linear Quadratic Regulator (LQR)

The optimal control problem minimizes:

$$J = \int_0^\infty (x^T Q x + u^T R u) \, dt$$

The optimal control law is:

$$u = -Kx$$

where K is obtained from the solution of the Algebraic Riccati Equation:

$$A^T P + PA - PBR^{-1}B^T P + Q = 0$$

and:

$$K = R^{-1}B^T P$$

Closed-Loop Observer-Controller System

Combining LQR and observer gives:

$$\begin{aligned} u &= -K\hat{x} \\ \dot{\hat{x}} &= A\hat{x} + Bu + L(y - C\hat{x}) \end{aligned}$$

The closed-loop system becomes:

$$\dot{x}_{cl} = \begin{bmatrix} A - BK & BK \\ LC & A - LC - BK \end{bmatrix} x_{cl}$$

Inverted Pendulum Model and Robustness

The framework is applied to the Furuta pendulum which is a nonlinear rotational inverted pendulum system that was linearized about equilibrium in the previous assignment. It serves as a benchmark for multivariable control design. Further, to model uncertainty, I introduced additive Gaussian noise:

$$\dot{x} = (A + \sigma.w(t))x + Bu$$

where:

$$w(t) \sim \mathcal{N}(0, \sigma^2 I)$$

This evaluates system robustness under stochastic perturbations. The reason the noise is included within the state matrix A is to observe the effects of noise that is correlated with the state only.

Observations

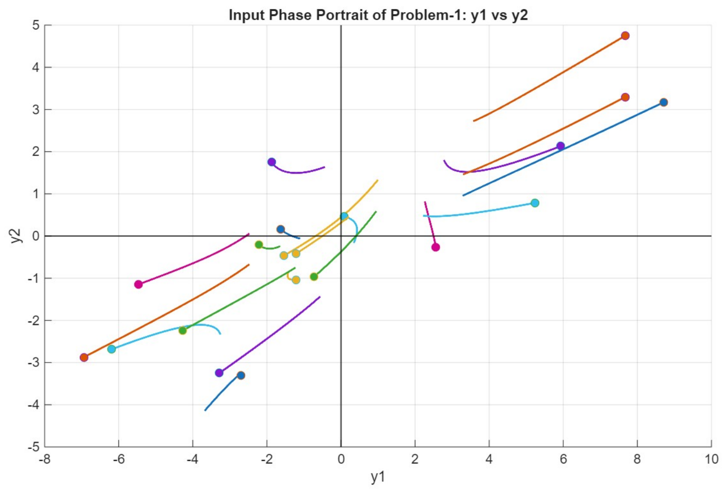


Figure 1: Input Phase portrait(y1 vs y2)-Problem 1

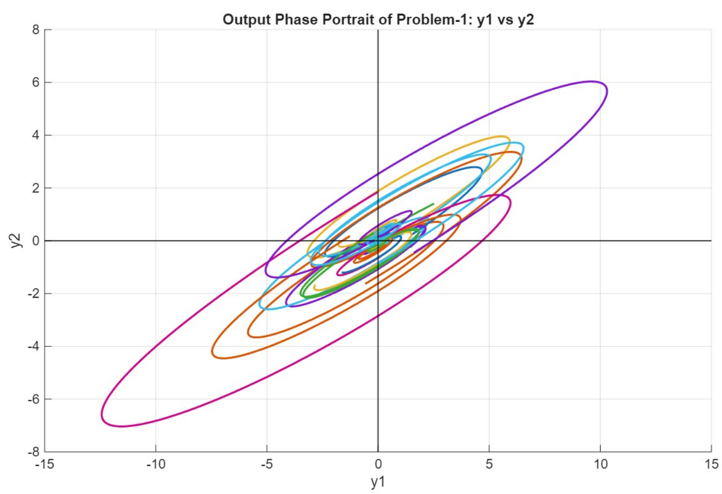


Figure 2: Output Phase portrait(y1 vs y2)-Problem 1

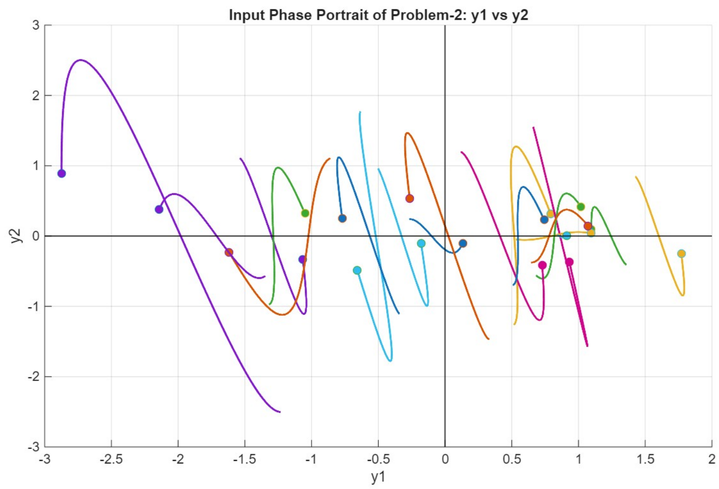


Figure 3: Input Phase portrait(y1 vs y2)-Problem 2

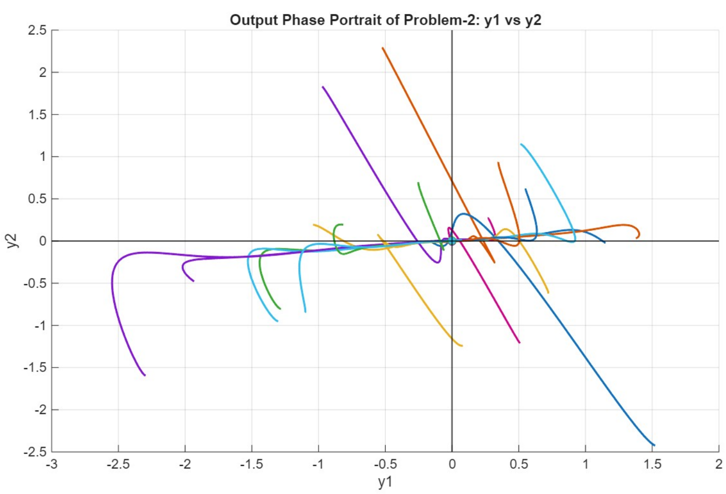


Figure 4: Output Phase portrait(y1 vs y2)-Problem 2

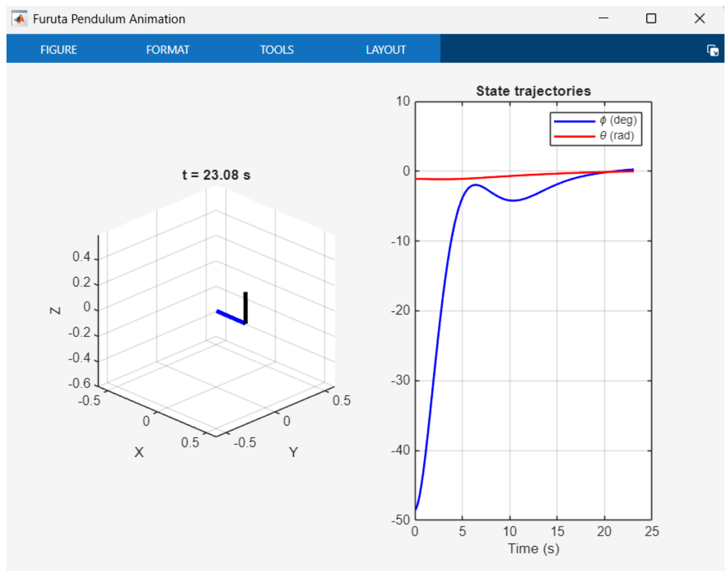


Figure 5: Furuta Pendulum Animation

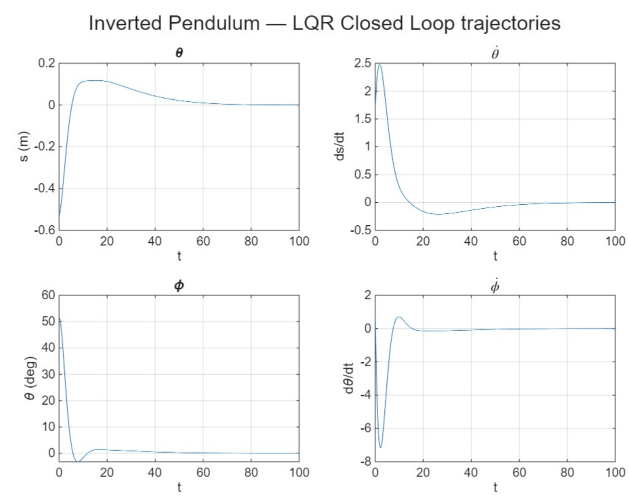


Figure 6: Plots of each of the state variables over the animation duration

Result

The developed state-space control framework successfully integrates system analysis, state estimation, and optimal feedback control into a unified closed-loop architecture.

The controllability and observability analysis confirmed that the system has the following dimensions:-

- $n_{V_{c\bar{o}}} = 4$; $n_{V_{co}} = 5$; $n_{V_{\bar{c}\bar{o}}} = 3$; $n_{V_{\bar{c}o}} = 2$

Based on these properties, Kalman decomposition enabled a structured separation of the state space into controllable and observable subspaces, ensuring that only dynamically relevant subspaces were retained for control design. Further, the similarity transformation was constructed.

A reduced-order Luenberger observer was successfully designed using the observability Gramian, enabling accurate reconstruction of unmeasured states. The estimation error dynamics were observed to be stable, confirming asymptotic convergence of the observer. This ensured that the controller operates on reliable state estimates even in the absence of full-state measurement.

The optimal state-feedback gain matrix K computed via the Linear Quadratic Regulator (LQR) approach and the solution of the Algebraic Ricatti equation resulted in a stable closed-loop system with significantly improved transient performance.

The combined observer-controller architecture demonstrated stable closed-loop behavior, as verified through eigenvalue analysis of the augmented system. All closed-loop poles were located in the left-half of the complex plane, confirming asymptotic stability.

The framework was further validated on the nonlinear Furuta pendulum system. The linearized LQR-based controller successfully stabilized the system about the equilibrium point, and the time-domain simulations along with animation confirmed smooth convergence of both angular and rotational states.

Finally, we introduced additive white Gaussian noise (AWGN) which provided several insights into robustness. The system maintained bounded responses under stochastic disturbances bounded by $\sigma \leq 0.1$, demonstrating that the designed controller-observer pair preserves stability and acceptable performance in the presence of moderate noise.

Overall, the experiment confirms that the integrated Kalman decomposition–observer–LQR framework provides a reliable and systematic methodology for controlling multivariable state-space systems, including nonlinear benchmarks under linearized operation.

Discussion

Amaey Advait(24EC10057):
In this assignment, I first analyzed the system in state-space form to study its controllability and observability and used Kalman decomposition to separate the controllable and observable subspace from the others. Based on this structure, I designed a reduced-order Luenberger observer to estimate only the detectable states with stable convergence.

I then implemented an optimal LQR controller to stabilize the closed loop system. Then, I closed the loop of closing the closed-loop system through simulation and the Furuta pendulum animation. Finally, I introduced additive noise to check robustness, and observed that the system maintained stability despite slight disturbances within a measurable threshold.

One of the aspects I found extremely fascinating was designing the Reduced Order Luenberger observer gain matrix by myself. It was challenging but also extremely satisfying to verify the working of that particular MATLAB function. It was also interesting to observe that the system was not nearly as robust as I had first thought. The closed loop poles are in the left half plane for both problems and relatively constant over multiple trials even with slight perturbation.

Through this experiment, I gained a deep understanding of the fundamental blocks used to synthesize a robust control which can be applied for any dynamical system that I model. Thank you for the opportunity.